

G8-DESIGN-AUDIT-v2: A/B/C mechanism comparison

ZFL-CX

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1 Frozen state

$G7 = \text{FROZEN}, G8\text{-DESIGN-AUDIT-v1} = \text{DONE}$	$G8\text{-01} = \text{KILLED}, G8\text{-02} = \text{KILLED}, G8\text{-03} = \text{KILLED}$
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$\mathcal{C}_{\text{PROVEN}} = \{C_E^{\text{weak}}, C_{L^3}, C_F^{\text{conc}}, C_I^{\text{freq}}, C_J^{\text{neg}}\}$ $\mathcal{C}_{\text{PROVEN}} \Rightarrow \perp \text{ OPEN}$ Pattern: static $\rightarrow \|u\|_3 \|\nabla \omega\|_2^2$,
dynamics $\rightarrow R_X$ supercrit, pressure $\rightarrow$ same scale no coercivity.

2 Correction

Do NOT assume $\xi^T S \xi > 0$. That is what must be proved. C_J^{neg} allows compressive alignment epochs. Any mechanism assuming $\alpha = \xi^T S \xi > 0 \Rightarrow \text{KILLED}$.

3 Candidate A: Scale dynamics

Need derived law, not postulated $r(t) \sim (T^* - t)^{1/2}$. Require: $r(t)$ law compatible with NS $\Rightarrow$ restriction on W_+ : $\int^{T^*} W_+ < \infty$ or $\perp$. Falsifiable claim: $\exists c > 0, T^* < \infty \Rightarrow \dot{r}/r \leq -cW_+$ or similar quantitative link. Kill if: $r(t)$ free choice of concentration sequence, no quantitative inequality, ends in $\|u\|_3 \|\nabla \omega\|_2^2$, or assumes regularity.

4 Candidate B: Blow-up profile

Rescaled limit $u^{(n)}(y, s) = r_n u(r_n y, r_n^2 s)$ with $C_F^{\text{conc}}, C_I^{\text{freq}}, C_J^{\text{neg}}$. Question: does nontrivial limit necessarily produce $\int \omega^T S \omega > 0$ in sufficient measure? Barrier: nontrivial profile $\nrightarrow$ positive stretching.
Need identity forcing sign, e.g. $\int \alpha |\omega|^2 \geq c_0 > 0$ for limit. Kill if: no identity forcing $\alpha > 0$, confuses concentration with stretching, only another necessary condition, or needs H^1 compactness equivalent to conclusion.

5 Candidate C: S - Ω interaction [Most interesting]

Vorticity: $(\omega \cdot \nabla)u = S\omega + \Omega\omega$, but $\omega^T \Omega \omega = 0$, so Ω no direct contribution to W . Needed: $\Omega\text{-dynamics} \Rightarrow$ restriction on $\dot{\xi} \Rightarrow \alpha = \xi^T S \xi \Rightarrow W_+$. Recall $D_t \xi = S\xi + \Omega\xi - \alpha\xi$, $\dot{\alpha} = \xi^T D_t S \xi + 2\xi^T S(D_t \xi)$. Ω enters via $\dot{\xi}$ rotation. Barrier E-07/E-08: controlling $D_t e_i$ requires $L^\infty S$ which we do not have. Falsifiable claim needed: e.g. Ω forces ξ away from compressive eigenvectors of S quantitatively, or $\int_0^{T^*} |\Omega\xi| < \infty$ under $\mathcal{C}_{\text{PROVEN}}$. Kill if: requires control of $D_t e_i$, assumes $\alpha > 0$, no quantitative inequality, ends in dimensional estimate, or only shows C_J^{neg} again.

6 Kill criteria v2

Route dies if: 1. scale not derived, 2. assumes $\alpha > 0$, 3. confuses concentration with stretching, 4. needs regularity equivalent to conclusion, 5. no quantitative inequality/identity, 6. ends again in $\|u\|_3 \|\nabla \omega\|_2^2$, 7. only another necessary condition without incompatibility. No automatic rescue via G8-02/03.

7 Decision rule

If A,B,C all fail falsifiable test: $G8\text{-DESIGN-v2} = CLOSED/BLOCKED$ Then conceptual class scale/profile/ $S\text{-}\Omega$ exhausted without fabricating progress. If one survives: it becomes sole authorized G8-04 with explicit mechanism, target, kill criteria.

8 Outcome TBD

This audit does not open G8-04. Its function is to decide which of A/B/C has falsifiable mathematical mechanism.